Using Therefore™ in Multi-Server and Multi-Tenant Environments

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PEOPLE PROCESS INFORMATION

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1. Introduction

What is covered by this document?

The purpose of this document is to explain how a Therefore™ system can be set up either as a multi-tenant system, or in multi-server environments. The difference between these two scenarios is explained in the first chapter. This document serves as a starting point for determining the best architecture approach in more complex Therefore™ installations. It also covers some considerations relating to clustering, licensing, and offering Therefore as a service, for example as an ASP.

This document refers to other Therefore™ white papers. If you are an end customer not affiliated with Canon, you can obtain these additional documents from your local Canon or Canon partner consultant.

What is not covered?

This document does not cover the specific technical details related to installing a Therefore™ system. This information can be found in the additional content to which this document refers.

Furthermore, this document does not take into account the needs or requirements relating to any specific country, region, or industry. Since regulations and design standards differ all over the world, the information in this documentation is meant purely as a general guideline. It is the responsibility of the person implementing the Therefore™ system to ensure the applicable requirements relating to data storage, security, and accessibility are met.

2. Multi-Server vs. Multi-Tenant: Understanding the Difference

This chapter will explain the difference between multi-server and multi-tenant systems, an oft-misunderstood distinction.

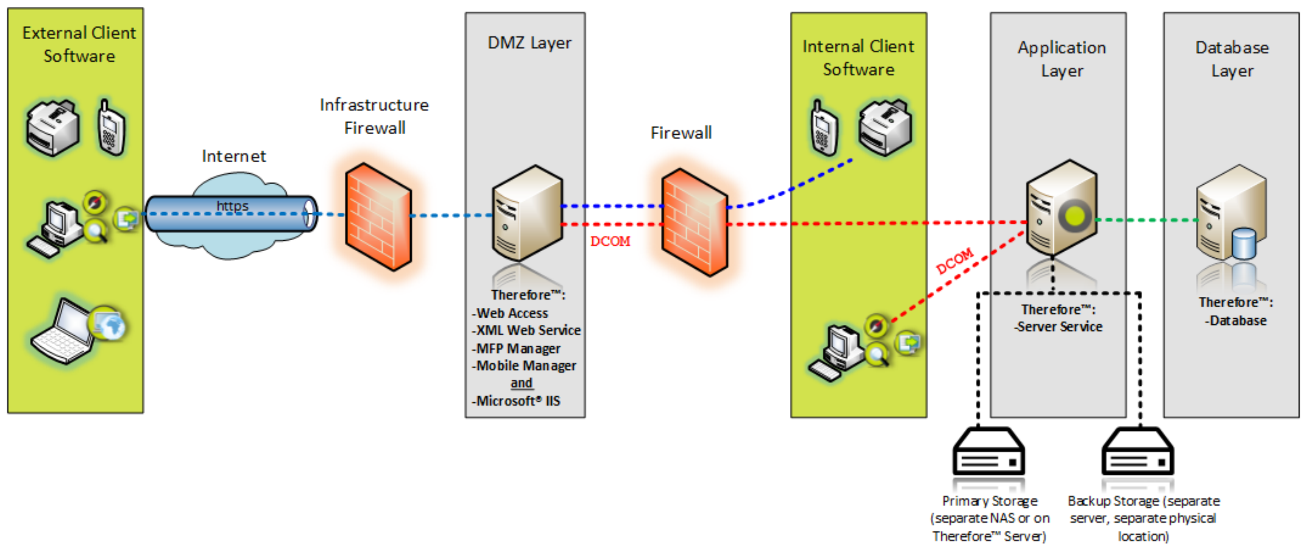
2.1 Multi-Server and Distributed-Server

First it is important to understand what is meant by multi-server and distributed-server environments.

Imagine a very simple Therefore™ system. The Therefore™ Server, database, storage, and all other services are all located on a single physical machine. Clients may connect via DCOM or XML Web Service. However, for a variety of reasons, it may be prudent to not have to rely on a single physical machine in one location. In this case, the different parts needed for a Therefore™ system will be distributed to different machines or layers. A distributed-server system will typically include three layers, as well as primary and backup storage:

1. DMZ Layer
2. Application Layer
3. Database Layer

The graphic below represents a typical Therefore™ installation on a distributed-server system.



In the image above, we see that there is only one Therefore™ Server in the application layer. "Multi-server" refers to an environment where more than one Therefore™ Server is being used. A multi-server environment is a type of distributed server environment. Continue reading for a more in-depth explanation of how multi-server systems work.

Detailed information on setting up a distributed server system can be found in the white paper **Best Practice for Distributed-Server Systems**.

2.2 Multi-Server Systems

Multi-server systems are typically deployed in large installations with multiple sites or locations. Customers with multiple branch offices around the world may be interested in increasing the speed at which documents can be retrieved, particularly if the network connection is slow or unreliable. In this situation, a multi-server configuration offers many advantages.

A multi-server environment can be configured with either a central or distributed database. In both cases, documents are replicated onto distributed Therefore™ Servers and thus retrieved locally. In the case of a central database, only the metadata is retrieved via the network. A network connection is hence always required. Where unstable network connections are an issue, availability of documents can be improved by also replicating the database onto the distributed servers. Even if the network connection between the servers is interrupted, users can get documents from all locally replicated data.

Customers who are seeking extra protection against hardware failure can use a failover cluster.

More details about licensing can be found in the **Licensing** section.

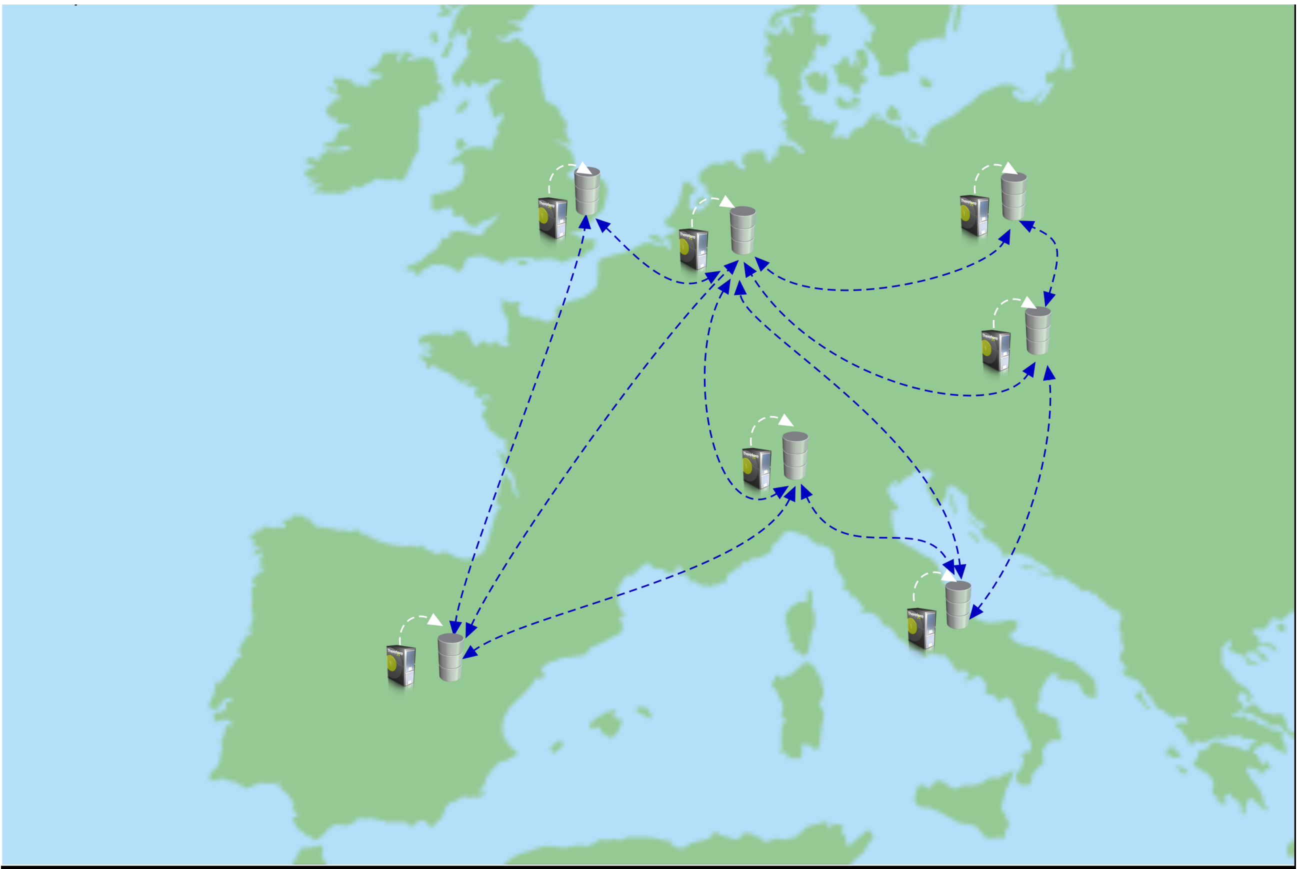
2.2.1 Configuration with a Central Database

A central database configuration allows for faster retrieval speeds for customers with limited bandwidth. In this setup, each location has its own Therefore™ Server with its own cache, but the database is centralized. Documents are replicated to the cache folder of the different servers at pre-configured times. Usually administrators choose to have the replication occur at night.

Therefore™ clients connect to their local Therefore™ Server and execute searches from there. Therefore™ Servers communicate with each other using DCOM and in case a document is not available locally, it is fetched from the correct server. All servers have the same rights, a buffer (for recently stored documents) and a cache (to temporarily store documents for faster retrieval). Primary and backup storage is defined on a system-wide level.

2.2.2 Configuration with a Distributed Database

This scenario is similar to the central database set-up, however each location has its own Therefore™ Server and also its own copy of the central database. The database is set up as a distributed database which replicates and synchronizes all data. Note that all databases must be maintained.



In the past, distributed database configurations were often used in areas with slow or unstable internet connections. Central database configurations have since become more popular due to more reliable network connections and the additional database maintenance costs associated with distributed databases.

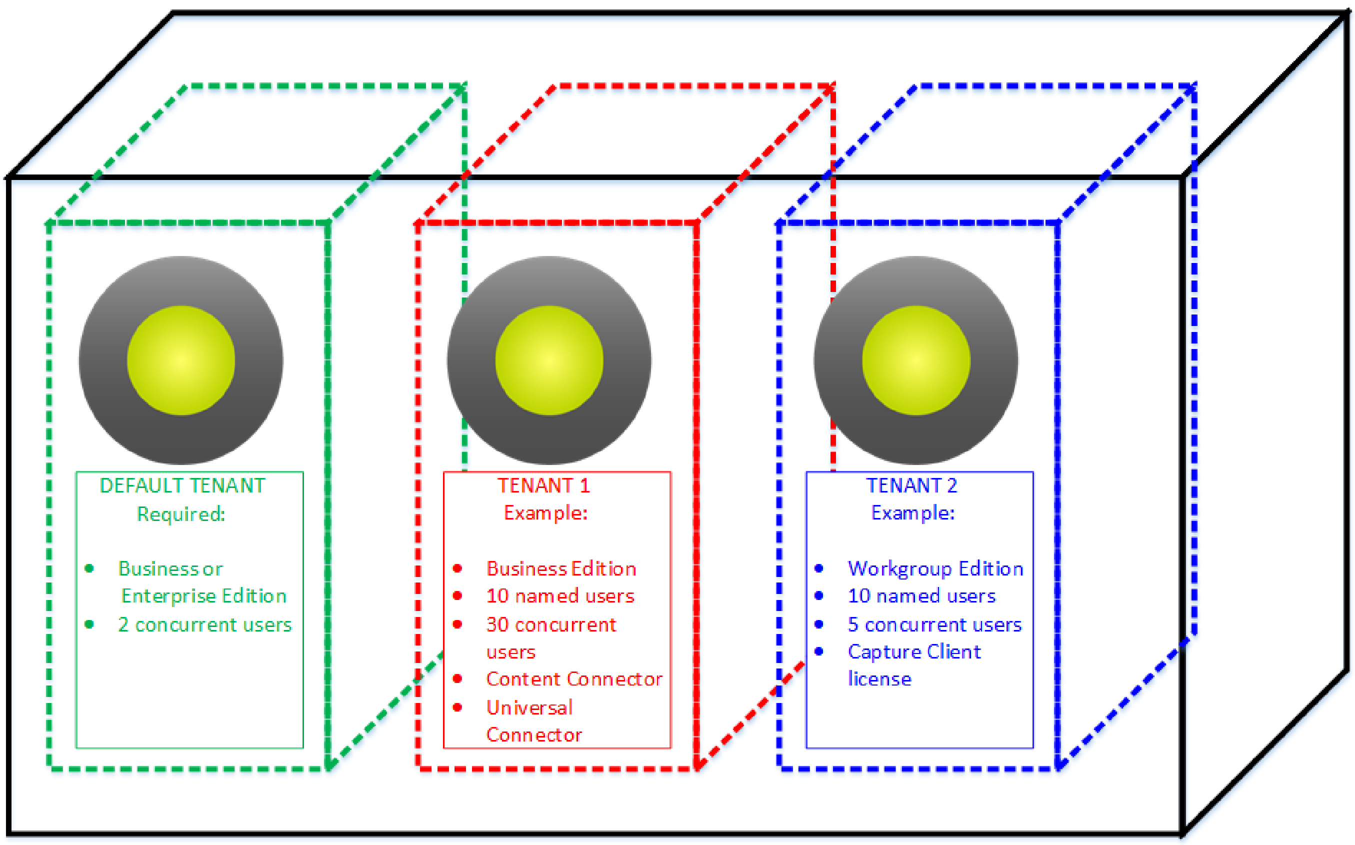
2.3 Multi-Tenant Systems

Multi-tenant systems are not the same as multi-server systems. A multi-tenant system refers to a single server which hosts multiple, separate Therefore™ systems. Using a building analogy, a single-server Therefore™ system would be a company in its own building. A multi-server system would be a company with branch offices in different buildings belonging to the company. A multi-tenant system is a large office building where many different companies can rent space and conduct their business separately.

Multi-tenancy allows **one** Therefore™ Server to host many tenants which have independent Therefore™ systems. This can be used within an organization to create totally independent Therefore™ systems for e.g. branch offices, or it can be used for hosted solutions (software as a service).

Multi-tenant systems require a default tenant before any other tenants can be created. The default tenant is used for system-wide administration and operational purposes, and may not be used for production as this may cause system instability. The default tenant requires its own license. The required configuration is **Business <u>or</u> Enterprise Edition (depending on region) with 2 concurrent users.**

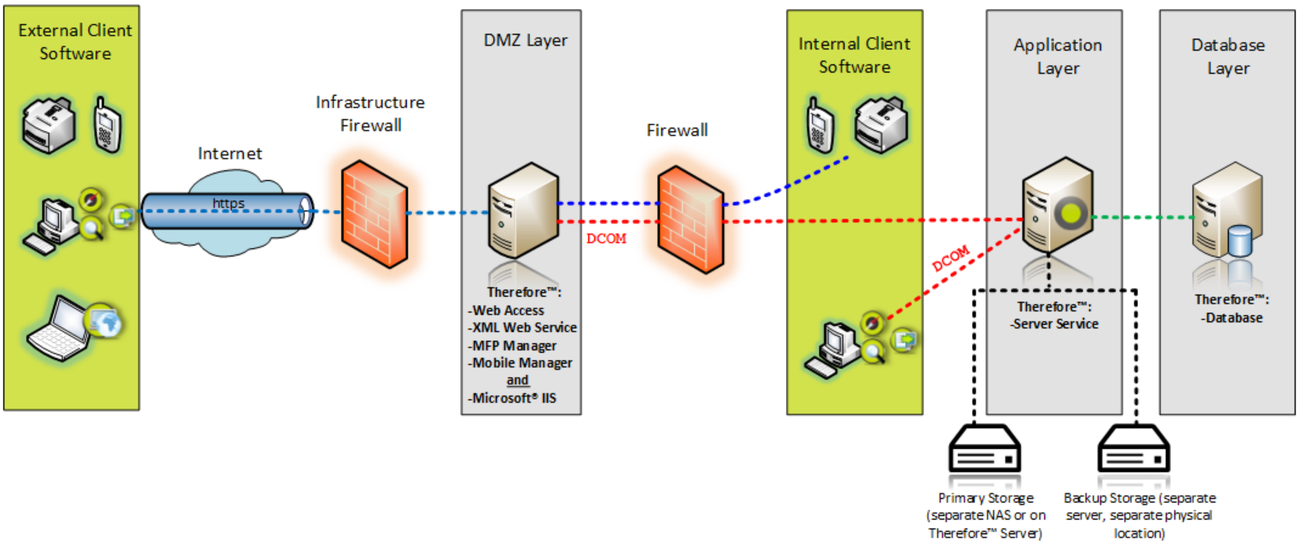
Each further tenant also requires its own, separate server license. Note that each tenant is a closed system, thus licenses cannot be shared across multiple tenants on a single system.



Complete technical details on creating a multiple-tenant system can be obtained from the **Multi-Tenancy Manual**. More licensing details can be found in the section on **Licensing**.

2.3.1 Service Infrastructure

The graphic below represents a typical setup of a multi-tenant system being used as a hosted service. The Therefore™ database is shown in a separate layer here, but this is not required. It could also be in the application layer, on the same machine as the Therefore™ Server.



3. Clustering and Failover

Setting up a failover cluster for the Therefore™ system is possible in environments where high uptime is business critical and data must always be accessible.

Clustering offers customers protection in case a server fails or goes offline. A cluster consists of 2 nodes (servers) and protects against hardware failure in the following way: When one node fails, another one jumps in automatically to provide those services in its place with no interruption for users. This process is called failover. Each failover server requires an additional Therefore™ Server license. More information about clustering can also be found on the Microsoft website.

Tutorials on how to set up a Therefore™ Server on a failover cluster can be found in the **Backup and Recovery** white paper.

4. Licensing

Concerning licensing, there is one major difference between multi-server and multi-tenant systems.

Since a multi-server environment is still a single system, only one main license is needed and user licenses can be distributed to all the servers. A server license is required for every server present in the system. For example, a 250-user system with 5 servers would require 5 server licenses. Each of the 5 servers must have user licenses assigned to them, for example 50 licenses each. Users can then draw from the pool of licenses assigned to the server they are currently connecting to. They may not draw a license from Server #1 when connecting to Server #2.

In a multi-tenant system, no licenses of any kind may be shared or distributed amongst tenants. Multi-tenant systems require a default tenant before any other tenants can be created. The default tenant is used for system-wide administration and operational purposes, and may not be used for production as this may cause system instability. The default tenant requires its own license. The required configuration is Business or Enterprise Edition (depending on region) with 1 with server license and 1 named user license.

Each further tenant also requires its own, separate server license, as well as user and connector licenses (if needed). Each tenant is a closed system, thus licenses cannot be shared across multiple tenants on a single multi-tenant system.

General information on Therefore™ Licenses can be found in the **Therefore™ Licenses White Paper.**

5. Therefore as an ASP

Therefore Corporation offers its information management systems in the cloud. This hosted SaaS solution is known as Therefore™ Online and is billed to customers as a subscription service. In essence, Therefore™ Online is a large multi-tenant system shared by hundreds of customers. Check with your local Canon or Therefore representative to find out if Therefore™ Online is available in your region.

Note: Canon Europe NSOs can find detailed technical information about the European hosted system in the Therefore™ Online PMD. This document can be obtained from the Therefore™ Product Manager at CEL.

Although Therefore™ Online is the recommended hosted solution, there may be multiple reasons why it may not be appropriate in a given situation:

- The customer is in a country where Therefore™ Online is not available.
- Local laws and/or regulations stipulate that data may not be stored in another country.
- The customer may not want to store their data offshore.
- The customer's data connection may be too slow to efficiently connect at long distances.

In these scenarios, it may be prudent to become an ASP and offer Therefore™ as a local hosted service by setting up a multi-tenant system and selling space to customers.

Advantages:

- Offer local storage for your customers. This can result in:
  - Compliance with local laws and regulations for your customers.
  - Faster document retrieval speeds.
- Provide a unique service your competitors may not be offering.
- Increasing ROI with each tenant/system sold, as costs do not increase.
- Easy to demo for customers once a demo tenant is set up.
- Shorter sales cycle and quicker deployment.
- Good source of recurring revenue.

However, keep the following considerations in mind:

- Since you own the hosted system, you are responsible for system maintenance and updates.
- Any downtime of your system will affect ALL your tenants.
- It is crucial to properly back up your storage and database. This would preferably be in a different physical location.
- Make sure to calculate any running costs for power or physical storage of your hardware.
- You are responsible for the security and integrity of your customers' stored data, as well as their connections to your system.
- Make sure your customers have enough monthly bandwidth to connect to your system.

6. Related Documents

- Backup and Recovery (white paper)
- Best Practice for Distributed Server Systems (white paper)
- Therefore™ Licenses (white paper)
- Therefore™ System Maintenance (white paper)
- Multi-tenancy manual (manual)